PRISM: M Document

Maintenance, Monitoring and Risks: PowerPulse Electricity Models

Part of PRISM	M = Maintenance, Monitoring and Risks (the fifth section of the PRISM structure)
Models covered	Model A: whole-home next-hour forecaster and spike detector (XGBoost). Model B: per-appliance hourly models for REFIT houses (Random Forest).
Source material	Electricity_Consumption_Forecasting_and_Spike_Detection.ipynb and PowerPulse_AllHouses_Colab.ipynb. The shared Colab link needs a Google sign-in and was not opened.
Evidence basis	Statements about the models come from the notebook code. Thresholds, cadences and owners marked "recommended" are suggestions, not taken from the notebooks. No run results were available.
Owner / version / date	

1. Purpose of This Document

This document tells the people who run these models what to look after, what to watch, and what can go wrong. It covers routine maintenance, ongoing monitoring, a risk register based on what the notebook code actually does, and a response guide for common problems.

2. System Snapshot

	Model A	Model B
Purpose	Hourly consumption forecast (7 and 30 days) with spike flags	Per-appliance hourly energy estimates for REFIT houses
Algorithm	XGBRegressor, 300 trees, depth 6, learning rate 0.05	RandomForestRegressor, 200 trees, min_samples_leaf 3, one per appliance
Input	hourly_power_consumption.csv (Datetime, Hourly_energy_kWh) in Google Drive	Cleaned hourly REFIT files in ./REFIT; appliance mapping for 20 houses (House14 not mapped)
Outputs	weekly and monthly forecast CSVs; weekly and monthly spike CSVs	all_houses_results.csv, summary_table.csv
Model saved to disk?	No. The model lives only in the Colab session.	No. Models are not stored; only metrics are saved.
Runs	Manually, top to bottom, in Colab	Manually, in a local or Colab notebook

3. Maintenance Plan

3.1 Routine tasks

Task	Recommended cadence	What to do
Refresh input data	Before each retrain	Model A: append the latest hourly readings to the CSV. Model B: add new or corrected REFIT files. Confirm column names and date ranges.
Retrain Model A	Monthly, or when a trigger in 3.2 fires	Run all cells; record MAE, RMSE and R2; compare with the previous run; save forecasts and spike CSVs under a dated name.
Retrain Model B	When data or appliances change	Run all cells; review summary_table.csv; check the count of appliances with MSE ratio under 2.0.
Archive outputs	Every run	Copy CSV outputs and metrics into a dated folder. The notebooks overwrite files of the same name.
Record environment	Every run	Note xgboost, scikit-learn, pandas and numpy versions. The Model A install cell is not version-pinned.
Review spike rule	Quarterly	Check the median + 3 x 1.4826 x MAD rule against recent genuine high-usage hours; adjust the multiplier if flags are too many or too few.

3.2 Retraining triggers

- Rolling forecast error is clearly above the baseline set at the last accepted run (see section 4).
- A change in the household: new appliance, occupancy change, solar or battery installation, tariff change.
- A gap or fault in the meter data that was later corrected.
- A seasonal change not yet represented in the training data. Model A only learns months it has seen.
- Library upgrades that change model behaviour.

3.3 Run steps: Model A

- Confirm hourly_power_consumption.csv is in MyDrive/Electricity_consumption with Datetime and Hourly_energy_kWh.
- Mount Drive and run cells in order. Cell 3 stops with an error if the file or columns are missing.
- Check the printed row count and date range for sense before trusting any output.
- Compare Cell 7 metrics with the baseline. Open the Cell 8 plot to confirm predictions follow actuals.
- Confirm all four CSV files were written in Cell 13 and archive them.

3.4 Run steps: Model B

- Place the house CSV files in ./REFIT and set DATA_PATH and OUTPUT_PATH.
- Check the Step 8 log: houses skipped for a missing mapping, loading failures and training failures are printed, not raised.
- Review the number of houses processed against the number of files supplied.
- Review summary_table.csv and the statistics block; archive both CSV files.

4. Monitoring Plan

The notebooks produce evaluation numbers only when they are run. Nothing monitors the models automatically. The checks below are recommended; the starting thresholds are suggestions to be replaced by values from the first accepted run.

4.1 Model A

Signal	How to measure	Recommended alert
Forecast accuracy	Once actuals arrive, compare the first 7 days of each forecast with actuals; compute MAE and compare with test MAE from Cell 7.	Rolling MAE more than about 25% above the baseline test MAE
Bias	Mean of (prediction minus actual) over the same window	Consistent over- or under-prediction across several days
Spike rate	Share of forecast hours flagged Potential Spike, per run	Share far above or below the rate seen at the accepted run
Spike usefulness	Of flagged hours, how many actuals later exceeded the same-hour threshold	Low hit rate for several runs: revisit the rule
Input quality	Rows loaded vs expected; missing hours; duplicate timestamps; unusual zeros	Any missing stretch; values outside plausible range
Long-horizon drift	Compare days 1 to 7 error with days 8 to 30 once actuals exist	Large error growth: shorten how far the monthly forecast is trusted

4.2 Model B

Signal	How to measure	Recommended alert
Generalisation	mse_ratio (test MSE divided by train MSE) per appliance	Ratio of 2.0 or more is flagged by the notebook; track the count over time
Fit quality	test_r2 per appliance; counts above 0.5 and 0.9	Any appliance that drops sharply from the previous run
Data coverage	Hours remaining after dropping missing and impossible hours	Large fall in usable hours for a house
Metered share	Printed Metered energy percentage per house	Big change between runs
Pipeline health	Houses processed vs files supplied; skipped and failed houses	Any unexpected skip or failure

5. Risk Register

Severity is a judgement of the likely effect on trust in the output if the risk is real. Items marked "verify" are concerns read from the code that should be confirmed by running the notebook.

ID	Risk	Evidence in the notebook	Severity	Mitigation
A1	Training and forecasting features may be timed differently (verify)	Training rows use lag_1 as the hour before the row while the target is the hour after the row; forecast code uses lag_1 as the hour before the predicted hour, and hour is the predicted hour.	High	Align feature construction in training with generate_forecast, then re-evaluate and compare scores.
A2	Errors grow over long horizons	Forecast feeds each prediction back as history for 168 or 720 steps; the notebook warns about accumulated error.	High	Treat the 30-day forecast as indicative; validate by horizon; consider direct multi-horizon models.
A3	Final forecast model never sees the latest 20% of data	The model is trained on the earliest 80% only and reused for forecasting.	Medium	After evaluating, retrain on all data before producing operational forecasts.
A4	Seasonality poorly learned	Month is a feature, but only months present in the data can be learned; no weather or holiday inputs.	Medium	Use at least a full year of data; consider weather and holiday features.
A5	Spike rule untested and has an edge case	Flags are never compared with actual events. If an hour's MAD is zero, the threshold equals the median.	Medium	Backtest flags on past data; set a minimum MAD floor.
A6	Gaps in the series break lag logic	Lags use row positions (iloc and shift); duplicates are averaged but missing hours are not filled.	Medium	Reindex to a continuous hourly grid and handle gaps explicitly.
A7	Single split gives limited confidence	One 80/20 chronological split.	Low	Use rolling-origin backtesting.
B1	Target leakage possible (verify)	Features are all columns except target kWh columns and house, so same-hour appliance power, max, min and std columns and the appliance's own lags may be included.	High	Print feature_cols; drop columns derived from the target appliance in the current hour; decide about own lags.
B2	Train scores are in-sample	Random forest train MSE is measured on training data; ratio threshold of 2.0 is a rule of thumb.	Medium	Use cross-validation by time blocks; judge mainly on test metrics.
B3	Cleaning not applied in run loop	clean_house is defined but Step 8 loads files assumed to be already cleaned.	Medium	Document how cleaned files were produced or call the cleaning step.
B4	Incomplete or uncertain appliance labels	Mappings include Unknown_2, Unknown_3, Unknown_5, Unknown_7; House14 absent; House12 skip list.	Medium	Label or exclude unknown appliances; document the missing house.

ID	Risk	Evidence in the notebook	Severity	Mitigation
B5	Failures are silent	Loading and training errors are caught and printed, then the loop continues.	Medium	Collect failures into a report and stop the run if too many occur.
B6	Hard-coded paths and mixed instructions	Headings mention Google Drive; code uses ./REFIT and ./PowerPulse_AllHouses_Results.	Low	Move paths to one configuration cell.
S1	Models not persisted	Neither notebook saves a trained model.	Medium	Save models with version and data range; keep the run log.
S2	Unpinned dependencies	Model A installs xgboost and ipywidgets without versions.	Low	Pin versions in a requirements note.
S3	Privacy of consumption data	Household readings can reveal routines; outputs are written to Drive or local folders.	Medium	Restrict folder access; do not share data with the notebook.
S4	Dependence on Google Drive state	Model A stops if the Drive folder or file is missing.	Low	Keep a backup copy of the dataset.

6. Response Guide

Symptom	Likely cause	Action
FileNotFoundError or missing columns (Model A)	Wrong Drive folder, file name or headers	Fix the path or headers; rerun from Cell 3.
Forecast is flat or unrealistic	Recursive drift, bad recent data, or feature mismatch (A1)	Inspect last observed data; apply the A1 check; retrain.
Very many or very few spikes	Threshold unsuitable after a pattern change, or MAD near zero	Review per-hour median and MAD; adjust the rule; retrain.
Model B R2 near 1.0 for many appliances	Leakage (B1)	Audit feature_cols and rerun.
Model B houses missing from results	No mapping, load error or training error	Read the Step 8 log; fix file names or mappings.
Metrics much worse after retraining	Data problem, changed household, library change	Compare data ranges and versions with the archived run; restore the previous setup if needed.

7. Review Checklist and Change Log

7.1 Before accepting a run

- Data range and row counts printed and sensible.
- Metrics recorded and compared with the previous accepted run.
- Plots inspected; no obvious flat lines or negative values.
- Outputs archived with date and library versions.
- Open risks reviewed, especially A1 and B1 until resolved.

7.2 Change log

Date	Version	Change and reason	Reviewed by